

Praveenraj Manickam

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PROFESSIONAL SUMMARY

Data Engineer with 3.5+ years of experience building scalable data and ML pipelines across telecom and wind energy domains. Experienced in processing large-scale SCADA time-series data using PySpark and PostgreSQL, implementing anomaly detection and forecasting models. Strong foundation in SQL, ETL workflows, distributed data processing, and ML lifecycle management using MLflow. MSc in Data Analytics (UK) with focus on data engineering and production-grade ML systems.

WORK EXPERIENCE

Data Engineer | Pixxon AI (Greentech Pvt Ltd) | Coimbatore, India January 2026 – Present

- Built ETL pipelines to process 10-minute SCADA data from 15+ wind turbines using Python and PostgreSQL..
- Developed LightGBM models for anomaly detection and turbine performance forecasting.
- Built FastAPI services to deliver real-time analytics and operational KPIs.
- Used MLflow and RabbitMQ for model tracking and scalable ML workflows.
- Optimized SQL queries and data processing pipelines to improve system performance, reliability, and real-time data availability.

Data Analyst | Trend Clothing | Birmingham, UK December 2024 – December 2025

- Analysed customer behaviour and sales trends to support product selection and market positioning.
- Built interactive Power BI dashboards for revenue and category performance tracking.
- Automated reporting processes using SQL and Power BI for faster decision-making.
- Achievement:** Improved client conversion by 30% through data-driven insights.

Network Data Analyst | Quadgen Wireless | Bangalore, India July 2022 – December 2023

- Processed and transformed large telecom datasets using SQL and PySpark for KPI reporting.
- Built automated ETL workflows and analytical datasets for network performance monitoring.
- Developed Power BI dashboards for uptime, latency, and service quality tracking.
- Achievement:** Improved reporting accuracy by 60% through optimized SQL logic and validation controls.

PROJECTS

HR Analytics Dashboard – Power BI Project July 2025

- Designed an interactive HR analytics dashboard using **Power BI**, **DAX**, and **Power Query** to analyse attrition patterns, salary slabs, age groups, and departmental KPIs, enabling faster data-driven HR decisions.

Coronary Heart Disease Risk Prediction – Machine Learning | Dissertation January 2025

- Built **XGBoost** and **Random Forest** models in **Python** to predict CHD risk using lifestyle features across India, USA, UK, and France, identifying smoking, obesity, and inactivity as major contributors.

EDUCATION

MSc in Data Analytics (Merit) | Aston University | Birmingham, UK January 2024 – July 2025

- Modules:** Machine Learning, Business Intelligence, Forecasting & Time Series, Data Engineering, Statistical Modelling, Applied Analytics.

BEng in Electronics & Communication Engineering (7.4/10) | Anna University | Chennai, India August 2018 – August 2022

SKILLS AND CERTIFICATIONS

- Technical Tools:** SQL | Python | PySpark | PostgreSQL | MLflow | RabbitMQ | Power BI | Streamlit | Git | JIRA.
- Data Skills:** ETL & Data Pipelines | Data Modeling (Star/Snowflake) | Time-Series Processing | Anomaly Detection | Event-Driven Architecture | Distributed Processing | Batch & Streaming Concepts.
- Analytics & ML:** Feature Engineering | Regression & Forecasting | Performance Evaluation (RMSE, R²) | Statistical Analysis
- Certifications:** Microsoft Power BI Data Analyst Associate (PL-300) | Python for Data Science (Coursera)

ADDITIONAL INFORMATION

Languages: English, Tamil, Malayalam. **Availability:** 15 days or less Notice Period